

The Wheel Atlas

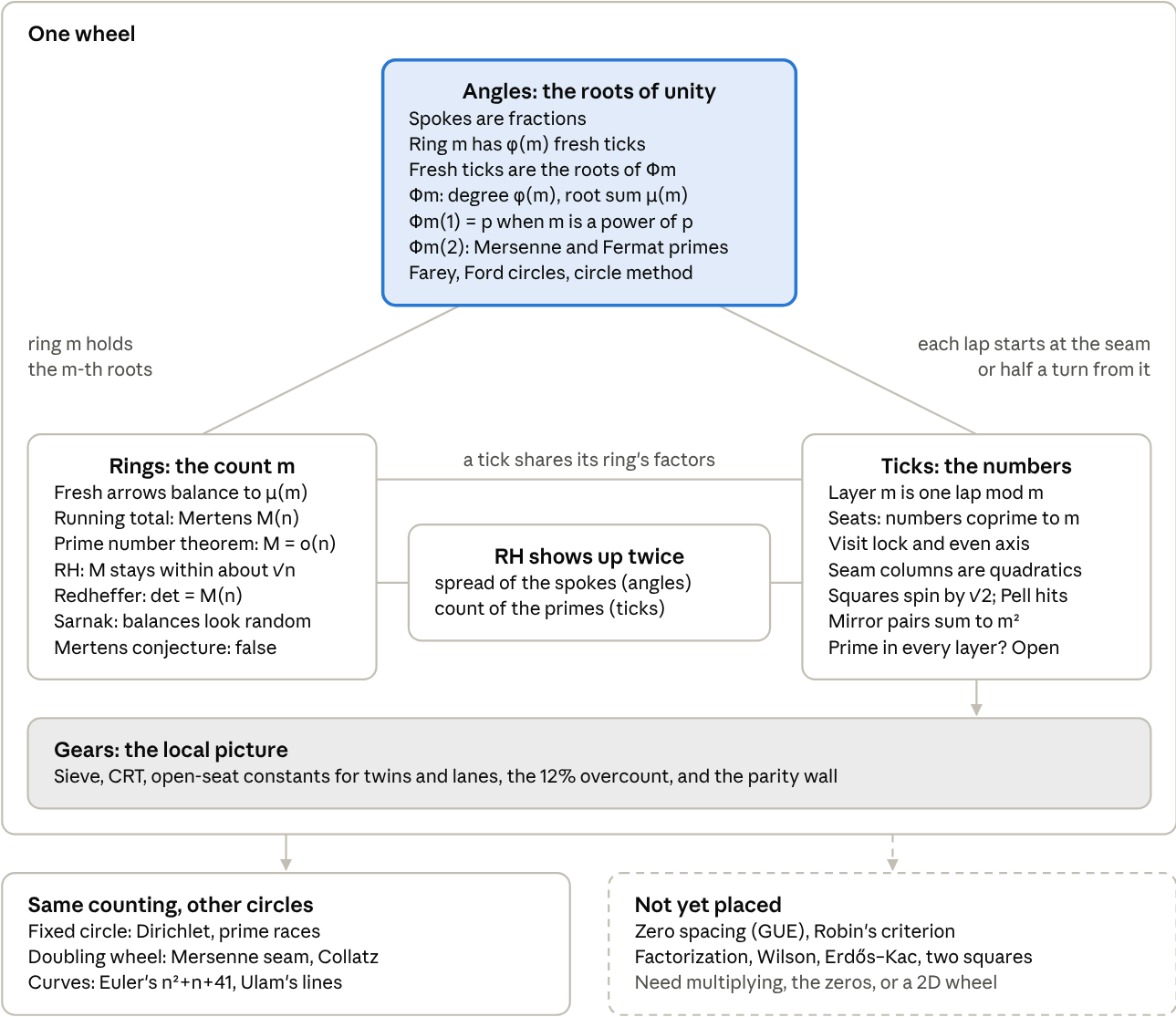
Oct 6, 2026 · @obi

At a glance

Everything placed so far lives in three worlds on one wheel, joined by four bridges, with the gears adding the local picture underneath.

One wheel holds three worlds of prime facts, joined by four bridges

Each family sits where it falls out of the counting rule



the wheel's three worlds and four bridges

The angles are the skeleton. Every ring is one lap of remainders that starts at the seam or half a turn from it, which is why the ticks inherit the angles' divisibility. The Riemann Hypothesis shows up on both sides.

The skeleton is forced

Counting onto Gauss's rings in order leaves exactly one possible wheel, and it's ours.

Start with the rings alone. Ring m is m evenly spaced points on a circle, one ring for every whole number. Mathematicians call these points the roots of unity, and since Gauss they have been the home of fractions and of the Möbius balance.

Now count onto them: 1, 2, 3 and on, one number per point. Fill the rings smallest first, and go around each ring starting just past the seam. There is exactly one way to do that: ring m receives the m numbers after the triangular number $T(m-1)$, which is our wheel.

So the wheel isn't one design among many. It is the counting order of a skeleton that was already there. What the counting adds is the tick side: visits that lock composites, the even/odd axis, and the lanes at the seam.

Change one piece

Only our wheel keeps all eight facts. Every other winding breaks at least two, checked on every number up to 3 million.

Winding	Spokes	Visit lock	Two per spoke	Even axis	Mertens	Seam composite	Rich lane	Aligned laps
Our wheel: sizes 1, 2, 3 ...	yes	yes	yes	yes	yes	yes	yes	yes
Even sizes 2, 4, 6 ...	yes	yes	yes	no	no	yes	yes	yes
Fixed circle of 12	no	yes	no	yes	no	yes	yes	yes
Skip repeat points	no	no repeats	no repeats	no	yes	no seam tick	yes	no
Same points, renumbered along diagonals	yes	no	no	no	yes	yes	yes	no
Every other layer counted backwards	yes	no	no	no	yes	no	no	no

Winding	Spokes	Visit lock	Two per spoke	Even axis	Mertens	Seam composite	Rich lane	Aligned laps
Odd sizes 1, 3, 5 ... (layers end at squares)	no	no	no	yes	no	yes	no	no
Doubling sizes 1, 2, 4, 8 ...	no	no	no	no	no	no	no	no

The columns: every fraction draws a line (spokes); third visits are never prime (visit lock); no spoke holds more than two primes; the half-turn tick is always even (even axis); the running balance equals the Mertens function; the closing tick is composite; the opening tick is at least 1.5 times as prime-rich as chance; every lap starts at the seam or half a turn from it (aligned laps).

The visit lock survives exactly where the laps are aligned. The triangular numbers do that for free, because $T(m-1)$ is always a multiple of half of m . The Mertens column needs every ring size once, in order. Only sizes 1, 2, 3 give both.

The fixed circle keeps the lock but loses the two-prime limit: there, one open spoke carried 54,271 primes, which is Dirichlet's theorem at work. The doubling wheel loses everything, including the seam, where its closing ticks are the Mersenne numbers.

New fits found while checking

Ten links between pieces turned up while placing the facts, each checked by computer here. Most are classical facts seen in a new place, and each says which.

- 1. Every layer is one lap of a fixed circle.** Layer m holds exactly one number of each remainder mod m . On odd layers each tick is its number's remainder; on even layers the remainders are rotated half a turn, because $T(m-1)$ is always a multiple of half of m . So the wheel stacks one lap of every fixed circle, and a layer's seats are simply its numbers that share no factor with m . *Simple; checked to layer 5,000.*
- 2. Each ring is one polynomial, and its values name primes.** Ring m 's fresh ticks are the roots of the cyclotomic polynomial Φ_m , whose degree is $\phi(m)$ and whose roots sum to $\mu(m)$. $\Phi_m(1)$ is the prime p when m is a power of p , and 1 otherwise, which is the von Mangoldt function behind the prime number theorem. At 2, the rings multiply out to $2^m - 1$, and every ring but 1 and 6 adds a brand-new prime factor (Zsigmondy). A ring can be drawn with compass and straightedge exactly when $\phi(m)$ is a power of 2, which is where the Fermat primes come in. *Classical; checked to ring 300.*

3. **The Riemann Hypothesis shows up on both sides.** On the angle side, the arrows of every fresh tick on rings 1 to n add up to exactly $M(n)$. RH says those spokes spread around the circle evenly to within about $\sqrt{n}$ (Franel–Landau). On the tick side, RH says the count of primes stays within about $\sqrt{x} \cdot \log x$ of its smooth estimate (von Koch). *Classical; checked to ring 1,600, where the spread error grows like $n^{0.36}$.*
4. **The seam is a catalog of quadratic lanes.** Every column near the seam is a quadratic. Columns a triangular number of ticks before the seam (0, 1, 3, 6, 10 ...) are always composite, because they hold sums of runs of consecutive numbers, which factor. Columns after the seam all stay alive, and quadratic residues set each one's richness, which is where reciprocity enters. The first (discriminant -7 , a Heegner number) runs 1.97 times richer than chance. The richest of the first 4,000 ($c = 586$) runs 3.32 times richer and, per odd value, ties Euler's $n^2 + n + 41$. *Simple; checked on layers 200 to 5,000, and on 100,000 layers per lane.*
5. **A quadratic draws a straight line here only when twice its lead coefficient is a square.** A sequence $At^2 + \dots$ turns $\sqrt{2A}$ of a turn per step on the wheel. When $2A$ is a perfect square it locks onto a straight row; otherwise it circles forever. So Euler's $n^2 + n + 41$, $n^2 + 1$ and Ulam's diagonals are curves on this wheel, and its straight rows are a different family. *Simple; checked on eight quadratics.*
6. **The squares ride a $\sqrt{2}$ spin, and Pell's equation is where they land on the seam.** n^2 sits at angle $\sqrt{2} \cdot n - 1/2$ (in turns) to within $0.09/n$. It lands exactly on the seam only at $n = 1, 6, 35, 204, 1189 \dots$, the square triangular numbers 1, 36, 1225, 41616 ..., which come from Pell's equation. *Classical numbers, new placement; checked to $n = 2$ million.*
7. **Layers with many small factors are richest per seat.** A seat on layer m holds a prime about $m/\phi(m)$ times as often as a random number its size. On layer 9,240 ($2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11$), 27% of its 1,920 seats hold primes; on the prime layer 7,919, under 6% of 7,918 do. *Checked on layers 1,000 to 10,000, to within 1%.*
8. **The sieve's local picture overcounts.** Multiplying the open-seat fraction of every wheel up to the square root predicts 6% more primes than exist at 100 million. The gap grows toward 12% ($2e^{-\gamma}$), so the local picture alone can't get the count right: the same wall the knockout test hit. *Classical (Mertens); checked to 10^8 .*
9. **Mirror pairs add up to the layer's square.** Tick k and tick $m - k$ of layer m always sum to m^2 , so every layer comes folded the way Goldbach's problem is: on layer 6, $17 + 19 = 36$. Neighboring seam numbers do the same: $T(m-1) + T(m) = m^2$. *Simple; checked to layer 3,000.*

10. Each spoke's two shots are chained. The second shot is twice the first plus b^2 , where b is the spoke's denominator: a cousin of the Sophie Germain pairs $(p, 2p + 1)$. Among spokes with denominators up to 2,000, 10,062 have both shots prime. *Simple; checked.*

The atlas

79 prime facts, theorems and mysteries, each placed where it lives on the wheel. 41 are native (they fall out of the rule), 32 translate (they can be said on the wheel, but it adds no structure), and 6 are foreign so far.

Rings and angles

Fact	Where it lives	How it shows up	Fit	Status
Fractions	Spokes	Each fraction a/b is a straight spoke, first visited on ring b	Native	Proven
Euler's $\phi(m)$	Ring m	The count of fresh ticks on ring m , and of its prime seats	Native	Proven
$\phi(d)$ summed over the divisors of m is m	Ring m	Each tick on ring m is a fresh tick of exactly one ring d that divides m	Native	Proven
Möbius function $\mu(m)$	Ring m	The fresh arrows of ring m sum to $\mu(m)$	Native	Proven
Ramanujan sums	Ring q	The fresh arrows spun n times; at $n = 1$ this is $\mu(q)$	Native	Proven
Cyclotomic polynomials	Ring m	Ring m 's fresh ticks are the roots of Φ_m : degree $\phi(m)$, root sum $\mu(m)$	Native	Proven

Fact	Where it lives	How it shows up	Fit	Status
Von Mangoldt function	Ring m	$\Phi_m(1)$ is p on rings that are powers of a prime p , and 1 otherwise	Native	Proven
Mersenne numbers	Rings at 2	$2^m - 1$ is the product of $\Phi_d(2)$ over the d dividing m , so a Mersenne prime needs m prime	Native	Proven
Zsigmondy's theorem	Rings at 2	Every ring except 1 and 6 brings $2^m - 1$ a brand-new prime factor	Native	Proven
Constructible polygons, Fermat primes	Ring m	Ring m can be drawn with compass and straightedge exactly when $\phi(m)$ is a power of 2; only five Fermat primes are known	Native	Proven
Decimal periods	Spokes $1/p$	$1/p$ repeats every d digits, where d is the first ring at which p divides $10^d - 1$	Translates	Proven
Fermat's little theorem	Rings at a	p divides $a^{p-1} - 1$, so p first shows up as a factor on a ring that divides $p - 1$	Translates	Proven
Artin's primitive root conjecture	Rings at 2	2 is a primitive root mod p when p first divides $2^d - 1$ at $d = p - 1$	Translates	Open
Farey sequences	Spokes	The spokes born on rings 1 to n , in order around the circle	Native	Proven

Fact	Where it lives	How it shows up	Fit	Status
Farey neighbors, mediants, Stern–Brocot tree	Spokes	Neighbors a/b and c/d satisfy $bc - ad = 1$; the next spoke born between them is the mediant	Native	Proven
Ford circles	Drum	The empty zone around each column: no other tick of ring m comes within $1/(bm)$ of a/b	Native	Proven
Dirichlet approximation, continued fractions	Slip	Every angle has near-miss spokes, and its best ones are its continued fraction	Native	Proven
Hurwitz's theorem, the golden ratio	Slip	The golden angle is the hardest angle to approximate with spokes	Translates	Proven
Three-gap theorem	Curves, fixed circles	Ticks at an irrational step leave at most three gap sizes	Translates	Proven
Circle method (Hardy–Littlewood)	Wedges	Major arcs are the wedges around low-denominator spokes; minor arcs are the shimmer between	Native	Proven
Weyl equidistribution	Curves	A quadratic whose $2A$ is not a square spins evenly around the wheel	Translates	Proven

The running balance and the Riemann Hypothesis

Fact	Where it lives	How it shows up	Fit	Status
Mertens function $M(n)$	Running balance	The running sum of the ring balances, which is also the sum of every fresh arrow on rings 1 to n	Native	Proven
Prime number theorem	Running balance	Equivalent to the balance growing slower than the ring count: $M(n) = o(n)$	Native	Proven
Riemann Hypothesis	Running balance	Equivalent to $M(n)$ staying within $n^{1/2 + \epsilon}$	Native	Open
Franel-Landau	Spokes	RH is equivalent to the spokes born on rings 1 to n spreading evenly to within about $\sqrt{n}$	Native	Open
Von Koch's form of RH	Ticks	RH is equivalent to the prime count staying within about $\sqrt{x} \cdot \log x$ of $\text{li}(x)$	Translates	Open
Mertens conjecture	Running balance	The guess that the balance never passes $\sqrt{n}$; disproved in 1985	Native	False
Redheffer matrix	Ring divisibility table	Which ring revisits which, plus a column of ones; its determinant is $M(n)$	Native	Proven
Sarnak's conjecture (Möbius randomness)	Ring balances	The balances should not correlate with any simple pattern	Translates	Open

Fact	Where it lives	How it shows up	Fit	Status
Pólya conjecture	Ring balances	The Liouville version of the lean; it first fails at 906,150,257	Translates	False
Robin's criterion	Nowhere yet	RH as a bound on divisor sums	Foreign	Open
Zeta zeros, explicit formula	Problem Wheels	Zeros as spinning wheels that add up to the prime count	Translates	Open
Montgomery's pair correlation	Nowhere yet	The zeros space out like eigenvalues of random matrices	Foreign	Open
Skewes's number	Ticks	The prime count crosses $\text{li}(x)$ infinitely often, but not below 10^{19}	Translates	Proven

Ticks: the counting numbers

Fact	Where it lives	How it shows up	Fit	Status
Floyd's triangle	Layers	Its rows are the layers	Native	Proven
Visit lock	Spokes	The g -th visit holds a multiple of g (odd g) or of $g/2$ (even g)	Native	Proven
Each layer is one lap mod m	Layers	The seats are the numbers coprime to m , exactly $\phi(m)$ of them	Native	Proven
Two shots per spoke, chained	Spokes	At most two primes per spoke; the second shot is $2 \times$ the first + b^2	Native	Proven

Fact	Where it lives	How it shows up	Fit	Status
Parity axis	Half-turn	Evenness is pinned to the half-turn on every layer	Native	Proven
Triangular numbers	Seam	The seam holds $T(m)$, and neighboring seam numbers sum to squares	Native	Proven
Mirror pairs	Layers	Ticks k and $m - k$ of layer m sum to m^2	Native	Proven
Gauss's Eureka theorem	Seam	Every number is a sum of three or fewer seam numbers	Translates	Proven
Polite numbers	Columns before the seam	Columns a triangular distance before the seam hold sums of consecutive runs, so they are always composite	Native	Proven
Lazy caterer primes	First column after the seam	Discriminant -7 ; never blocked by wheels that are $3, 5$ or $6 \pmod{7}$; 1.97 times chance	Native	Open
Quadratic reciprocity	Seam columns	Turns "is $-D$ a square mod p " into "what is $p \pmod{D}$ ", which picks each column's blocking wheels	Native	Proven
Heegner numbers, Euler's $n^2 + n + 41$	Seam columns, curves	The first column's -7 is a Heegner number; Euler's polynomial is a curve here; column 586 ties it per odd value	Translates	Checked

Fact	Where it lives	How it shows up	Fit	Status
Bateman–Horn conjecture	Every straight row	A row's prime density is its open-seat product, matched within 0.3% on seven columns	Native	Open
Bunyakovsky conjecture	Every live row	Each live row should hold endless primes; no quadratic case is proven	Native	Open
Landau's fourth problem ($n^2 + 1$)	Curves	A curve on this wheel, since $2A = 2$	Translates	Open
Ulam spiral	Curves	Its diagonals ($4n^2 + bn + c$) are curves here; each winding straightens its own family	Translates	Checked
Sacks spiral	Seam	Sacks puts one square on each turn, all on one ray; the wheel puts one triangular number on each turn, all on the seam	Translates	Proven
Pell's equation, square triangular numbers	Squares' curve	The squares spin by $\sqrt{2}$ per step and hit the seam at Pell solutions	Native	Proven
Prime in every layer	Layers	A prime between neighboring triangular numbers (OEIS A066888)	Native	Open

Fact	Where it lives	How it shows up	Fit	Status
Legendre's conjecture	Squares' curve	A prime between neighboring squares, about $\sqrt{2}$ layers apart; open even assuming RH	Translates	Open
Oppermann's and Andrica's conjectures	Layers	Andrica: consecutive primes are always less than about $\sqrt{2}$ rings apart	Translates	Open
Bertrand's postulate	Rings	A prime in every band from radius r to $\sqrt{2} \cdot r$	Translates	Proven
Prime gaps	Layers	Cramér's guess would put a prime in every large layer; the proven $x^{0.525}$ and RH's $\sqrt{x} \cdot \log x$ are too coarse	Translates	Open
Seats by layer	Layers	A seat holds a prime $m/\phi(m)$ times as often as chance	Native	Checked

Gears: the local picture

Fact	Where it lives	How it shows up	Fit	Status
Sieve of Eratosthenes	Gears	Each wheel p takes out every p -th number	Native	Proven
Chinese remainder theorem	Gears	The wheels mesh independently	Translates	Proven
Mertens' third theorem	Gears	The open-seat product up to $\sqrt{x}$ overcounts primes, heading to $2e^{-\gamma} \approx 1.12$	Native	Proven

Fact	Where it lives	How it shows up	Fit	Status
Parity problem	The wall	Sieves can't tell an odd count of prime factors from an even one: the knockout wall	Native	Proven
Twin primes, twin prime constant	Neighbor seats	Seat pairs two ticks apart; the constant is the product of the open pair fractions	Native	Open
Prime k-tuples	Gears	A pattern is admissible when no wheel blocks all of its seats	Native	Open
Bounded gaps (Zhang, Maynard)	Neighbor seats	Infinitely many prime pairs at most 246 apart	Translates	Proven
Brun's theorem	Neighbor seats	The reciprocals of the twin primes add up to a finite sum	Translates	Proven
Chen's theorem	Neighbor seats	Infinitely many p with $p + 2$ prime or a product of two primes	Translates	Proven
Sophie Germain primes	Spokes	Chained shots $(p, 2p + b^2)$ are a cousin of $(p, 2p + 1)$	Translates	Open

Other windings, and what doesn't fit yet

Fact	Where it lives	How it shows up	Fit	Status
Dirichlet's theorem	Fixed circle, and every layer	Each open spoke of a fixed circle carries endless primes, shared evenly	Translates	Proven

Fact	Where it lives	How it shows up	Fit	Status
Chebyshev's bias, prime races	Fixed circle of 4	Primes that are 3 mod 4 usually lead; the size of the lead is known only under RH-type guesses	Translates	Open
Green–Tao theorem	Fixed circles	Some fixed circle has a spoke with any number of primes in a row	Translates	Proven
Mersenne seam	Doubling wheel	The doubling wheel's closing ticks are the Mersenne numbers	Translates	Proven
Collatz	Doubling wheel	Explored on Problem Wheels' doubling wheel	Translates	Open
Goldbach's conjecture	Mirror pairs, folded wheel	Each layer folds onto m^2 ; the local constraints show, the global claim is open	Translates	Open
Ternary Goldbach	Wedges	Proved with the circle method (Helfgott, 2013; widely accepted)	Translates	Proven
Unique factorization	Nowhere yet	The wheel shows only the factors a number shares with its own layer	Foreign	Proven
Wilson's theorem	Nowhere yet	A product over a whole fixed circle; the wheel adds, it doesn't multiply	Foreign	Proven
Sums of two squares	Fixed circle of 4	The 4-circle says which primes qualify; the two squares need a second dimension	Foreign	Proven
Erdős–Kac theorem	Nowhere yet	The count of prime factors is bell-shaped	Foreign	Proven

What doesn't fit yet

The six foreign pieces share one reason: the wheel adds, and they need multiplying, a second dimension, or the zeros themselves.

- **Multiplying.** Unique factorization, Wilson's theorem and Erdős-Kac are about whole products. The wheel only shows the factors a number shares with its own layer, never its full factorization. A home for them would need the gears running alongside, each number lit on every gear that divides it.
- **A second dimension.** Sums of two squares live on a flat grid of points, not on a line wound around circles. The 4-circle sorts which primes qualify, but the two squares need a wheel in two dimensions.
- **The zeros.** Montgomery's pair correlation and Robin's criterion talk about the zeta zeros and divisor sums directly. Problem Wheels draws the zeros as spinning wheels, but their spacing has no picture here yet.

The big open problems (Goldbach, twin primes, a prime in every layer, Legendre) do fit, but only locally. The wheel shows every constraint each gear puts on a seat. What's missing is always the last step, from "this seat is open" to "this seat holds a prime", which is the wall.

What to probe next

Five probes would test the map rather than decorate it.

- **Push the seam catalog.** Search columns past 4,000, and around the other spokes, for the richest lanes. Then check whether their richness tracks class numbers the way Euler's polynomial does.
- **Count the double shots.** Predict from the open-seat products how many spokes should get both shots prime, write the prediction down, then count.
- **Watch the Riemann Hypothesis from both sides.** Draw the spoke spread (angles) and the prime count's error (ticks) on the same wheel, and see whether they move together ring by ring.
- **Build the second dimension.** A wheel for points on a flat grid would give sums of two squares and the Gaussian primes a home.
- **Train a small network on wheel coordinates,** then compare the geometry it builds with this map.

The scripts behind every check here are `knockout_rule.py` and `atlas_checks.py`, alongside the other probes.

Sources

The pages behind the classical claims, each opened while building this.

- Farey sequences and Franel–Landau: [Farey sequence](#), [Farey neighbors](#), [Stern–Brocot tree](#), [Ford circles](#)
- Rings as polynomials: [cyclotomic polynomial](#), [product over divisors](#), [values at 1 \(OEIS A020500\)](#), [Möbius as a root sum \(OEIS A008683\)](#), [Ramanujan's sum](#), [totient sum](#)
- Values at 2 and 10: [Zsigmondy's theorem](#), [Mersenne primes](#), [Fermat numbers](#), [constructible polygons](#), [repeating decimals](#), [Artin's conjecture](#)
- The balance and RH: [Mertens function](#), [Möbius function and the prime number theorem](#), [Redheffer matrix](#), [von Koch's form](#), [Robin's theorem](#), [Pólya conjecture](#), [Liouville function](#), [Sarnak's conjecture](#), [Skewes's number](#), [li\(x\) above \$\pi\(x\)\$ to \$10^{19}\$](#) , [Montgomery's pair correlation](#)
- Approximation: [Hurwitz's theorem](#), [three-gap theorem](#), [circle method](#)
- Quadratics and the seam: [Floyd's triangle](#), [visit divisibility \(Math::PlanePath\)](#), [lazy caterer primes \(OEIS A055469\)](#), [Bateman–Horn conjecture](#), [Bunyakovsky conjecture](#), [Heegner numbers](#), [quadratic reciprocity](#), [Ulam and Sacks spirals](#), [square triangular numbers](#), [Gauss's Eureka theorem](#), [polite numbers](#)
- Gaps and layers: [prime in every layer \(OEIS A066888\)](#), [Landau's problems](#), [Legendre is open even under RH](#), [Oppermann's conjecture](#), [Andrica's conjecture](#), [Cramér's conjecture and the RH gap bound](#), [Baker–Harman–Pintz](#), [prime gaps and bounded gaps](#)
- Sieves and pairs: [Mertens' third theorem](#), [the sieve's \$2e^{-\gamma}\$ overcount](#), [parity problem](#), [prime k-tuples](#), [twin prime constant](#), [Brun's theorem](#), [Chen's theorem](#), [Helfgott and ternary Goldbach](#)
- Fixed circles and the rest: [Dirichlet's theorem](#), [Chebyshev's bias](#), [Green–Tao theorem](#), [sums of two squares](#), [Wilson's theorem](#), [Erdős–Kac theorem](#)